

Resolving Wigner Negativity, Semiclassical Caustics, and Schrödinger Nodes via the Uncertainty Principle

Brian S. Mulloy*

(Dated: 2026-05-01)

Wigner negativity, semiclassical caustics, and Schrödinger nodes follow from ignoring Heisenberg constraints — equivalently, from assuming unbounded classical action. Enforcing the action capacity of a physical system via symplectic phase-space cells resolves all three simultaneously. The resulting phase-space densities are provably non-negative, free of turning-point divergences, and strictly positive everywhere.

The WKB probability density [1, 2] diverges at every classical turning point. The Wigner–Weyl distribution [3] takes negative values for any state beyond the ground state. The exact wavefunction $|\psi_n(q)|^2$ is exactly zero at every node. Each is treated as a feature of its own formalism. They share a kinematic root. Each requires evaluating a density at a point in phase space, and the uncertainty principle does not permit that. A point is a region of zero phase-space area. No physical instrument resolves zero area, and no physical state is supported there.

The matrix of pathologies and textbook remedies is small. In the semiclassical formalism, the WKB density diverges at turning points and the Airy uniform density [1, 2] is the textbook remedy: connection formulas patch the wavefunction across each turning point in an asymptotic expansion local in q . The patched wavefunction, Wigner-transformed, is again negative. In the wave formalism, the Wigner function goes negative under interference and the Husimi Q -function [4] is the textbook remedy: convolution with a fixed coherent-state kernel of unit width gives a non-negative density at the cost of every scale below the ground-state cell. Schrödinger nodes [5, 6] are exact zeros of the wavefunction. They have no textbook remedy and are read as a structural feature of the spectrum. None of the three remedies addresses the others.

The three pathologies dissolve together when the question is asked at the resolution physics permits. The smallest cell of phase space that admits a physical answer has area $h/2$. Inside any larger cell, the conjugate substructure has reciprocal widths $\delta q = \hbar/\Delta p$ and $\delta p = \hbar/\Delta q$ [9]. The minimum area cannot be evaded by squeezing along any conjugate axis pair [10]. Together these fix a unique pair of squeezed Gaussian kernels of phase-space area $h/2$ at every point in the orbit’s covariance ellipse. The convolution of any quantum or classical phase-space density with such a kernel is non-negative [11] and finite. The same kinematic constraint that rules out pointwise infinity rules out pointwise zero, so the resolved density is strictly positive throughout phase space.

The Fermi blob [8] of an energy eigenstate sits on the classical orbit at the Bohr–Sommerfeld energy E_n . The orbit’s time-averaged second moments give classical semi-

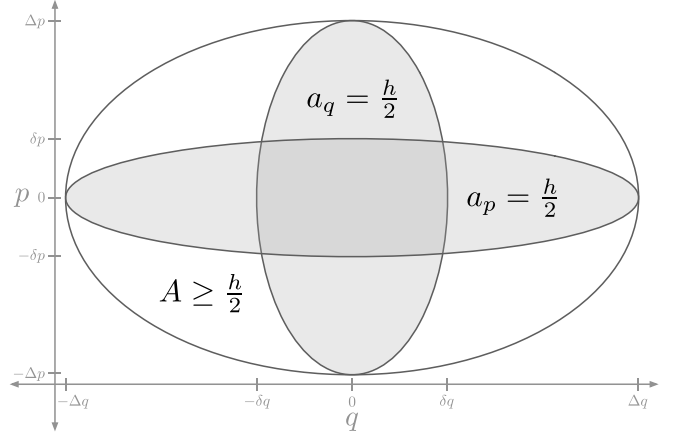


FIG. 1. **The symplectic quantum of action** (schematic, 1D case). The Heisenberg limit $\sigma_q\sigma_p \geq \hbar/2$ admits classical semi-axes $\Delta q = \sqrt{2}\sigma_q$, $\Delta p = \sqrt{2}\sigma_p$; the 1σ contour of the state’s covariance projects onto a Fermi blob [8] of phase-space area $A = \pi\Delta q\Delta p \geq h/2$. Inside the blob sit a conjugate pair of orthogonal quantum blobs, $a_q = \pi\delta q\Delta p = h/2$ and $a_p = \pi\Delta q\delta p = h/2$, summing to the quantum of action $h = a_q + a_p$. Their minor semi-axes are the reciprocal scales $\delta q = \hbar/\Delta p$ and $\delta p = \hbar/\Delta q$; a_q is squeezed along position, a_p along momentum. As A grows, a_q and a_p compress along their conjugate axes and phase-space resolution sharpens.

axes $\Delta q = \sqrt{2\langle(q - \langle q \rangle)^2\rangle_{\text{orbit}}}$ and $\Delta p = \sqrt{2\langle p^2 \rangle_{\text{orbit}}}$, and the blob has area

$$A = \pi\Delta q\Delta p \geq \frac{h}{2}. \quad (1)$$

The Heisenberg limit imposes $A \geq h/2$, with the lower bound saturated by minimum-uncertainty Gaussians. The Robertson–Schrödinger inequality [5, 6] is the constraint these moments respect, not their source. The reciprocal scales partition the blob’s interior; the conjugate quantum blobs [7?],

$$a_q = \pi\delta q\Delta p = \frac{h}{2}, \quad a_p = \pi\Delta q\delta p = \frac{h}{2}, \quad (2)$$

each carry the irreducible area $h/2$, a_q squeezed along the position axis, a_p along the momentum axis (Fig. 1). A non-eigenstate has no orbit; its moments come from the Robertson–Schrödinger covariance of the state directly.

The action capacity A and the Bohr–Sommerfeld action $A_{\text{BS}} = \oint p dq$ are different objects. Both have units of action; they describe different geometric content. For the harmonic oscillator they coincide and the Fermi blob equals the energy-shell ellipse. For anharmonic potentials they separate, with $A \leq A_{\text{BS}}$. The kernel widths δq and δp track A , the orbit’s covariance, not A_{BS} , the area it encloses.

The kernels of phase-space area $h/2$ corresponding to the conjugate quantum blobs are

$$G_{\delta q}(q, p) = \frac{1}{\pi} \exp\left(-\frac{q^2}{\delta q^2} - \frac{p^2}{\Delta p^2}\right), \quad (3)$$

$$G_{\delta p}(q, p) = \frac{1}{\pi} \exp\left(-\frac{q^2}{\Delta q^2} - \frac{p^2}{\delta p^2}\right). \quad (4)$$

$G_{\delta q}$ is squeezed along the position axis and spans Δp in momentum; $G_{\delta p}$ is the conjugate. Each saturates the symplectic-capacity bound at $h/2$. The same kernel, fixed by the orbit, applies to two distinct inputs. For the classical microcanonical density $W_{\text{cl}}(q, p) \propto \delta(H - E)$, the marginal of the convolution over p is

$$\rho_{\delta q}(q) = \int (W_{\text{cl}} * G_{\delta q})(q, p) dp, \quad (5)$$

finite even where P_{WKB} diverges. For the quantum Wigner function the convolved phase-space density is

$$P_{\delta q}(q, p) = (W * G_{\delta q})(q, p), \quad (6)$$

non-negative everywhere because the kernel attains the symplectic positivity condition [11]. The kernel’s Gaussian tails are everywhere positive, so both convolutions are everywhere strictly positive — a fact that returns when tunneling appears in the double-well barrier interior.

For the Morse oscillator $V(q) = D_e(1 - e^{-\alpha q})^2$ with $D_e = 12.5$, $\alpha = 0.2$, $m = 1$, the natural frequency $\omega_0 = \alpha\sqrt{2D_e/m} = 1$. Bound states have $A < A_{\text{BS}}$, with the gap widening as the orbit reaches into the soft-wall region. Figure 2 shows the semiclassical pipeline at $n = 0, 8, 16$. The WKB density carries the canonical Bohr–Sommerfeld interference pattern — the half-orbit phase advances by $\pi(n+1/2)$ and $\cos^2$ of that phase has n zeros and $n+1$ peaks — together with caustic divergences at the turning points. At resolution δq , the divergences become finite peaks and the lobe envelope persists. The Airy uniform density agrees on lobe count and envelope but has taller peaks and zero troughs because Airy is evaluated at infinite resolution. The resolved density is the same content sampled at the orbit’s natural scale $\delta q = h/\Delta p$.

The Morse ground state ($n = 0$) reveals a structural fact about the cells. The Bohr–Sommerfeld classical orbit reads $\Delta q \Delta p < h/2$ at low quantum numbers in anharmonic potentials [12]. The orbit can do this because it is a classical object, not a quantum state. The cells cannot. The smallest cell physics admits has area $h/2$, no

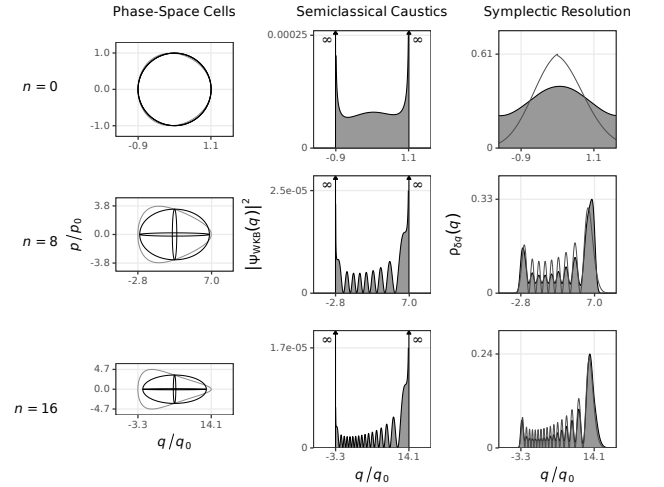


FIG. 2. **Morse oscillator, semiclassical.** Rows show eigenstates $n = 0, 8, 16$. *Left:* Classical orbit at E_n with the symplectic quantum of action overlaid. The orbit deforms from a circle ($n = 0$) to a horseshoe ($n = 16$) as it explores the soft-wall region. *Center:* Oscillating WKB density $|\psi_{\text{WKB}}(q)|^2 = (A^2/p) \cos^2(S/\hbar - \pi/4)$, with $n+1$ interference lobes between the turning points and divergences (rendered as infinity arrows) where $p \rightarrow 0$. *Right:* Resolved density $\rho_{\delta q}(q)$ with the Airy uniform density [2] overlaid. The $n+1$ lobes survive; the divergences become finite peaks. The Airy curve has higher peaks and deeper troughs at every n because it is evaluated at infinite resolution; the resolved density is the same content sampled at the orbit’s natural scale $\delta q = h/\Delta p$.

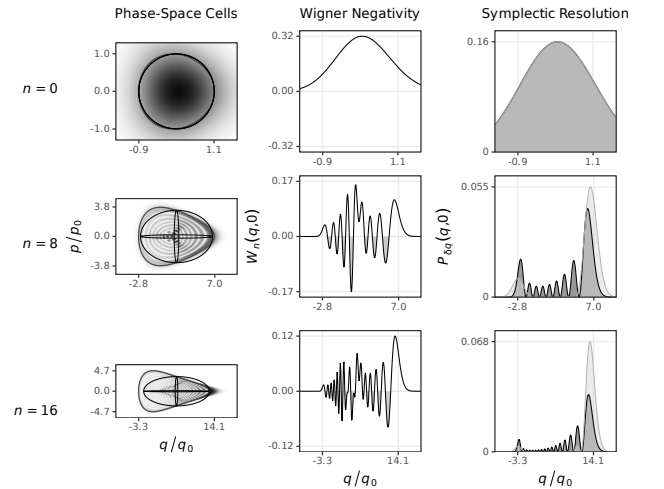


FIG. 3. **Morse oscillator, Wigner.** Same eigenstates as Fig. 2; same kernel widths from the same orbit. *Left:* Wigner distribution $W_n(q, p)$ with the symplectic quantum of action overlaid. *Center:* Cross-section $W_n(q, 0)$ showing the growing oscillation structure; negative regions deepen with n . *Right:* Cross-section $P_{\delta q}(q, 0)$ with the Husimi cross-section $Q(q, 0)$ overlaid. The Husimi kernel of width $q_0 = 1$ exceeds δq for $n \geq 8$ and collapses the multi-lobe structure into a single peak; δq tracks the orbit and resolves the lobes.

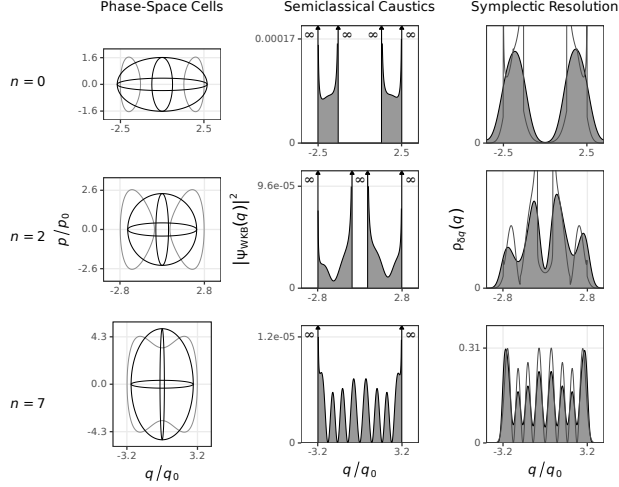


FIG. 4. **Symmetric double well, semiclassical.** Rows show $n = 0$ (lower symmetric doublet partner), $n = 2$ (upper symmetric doublet partner, sub-barrier), and $n = 7$ (above-barrier connected orbit). *Left:* Classical orbit at E_n with the symplectic quantum of action overlaid. Sub-barrier states have two disconnected orbital loops; the above-barrier state has a single peanut. *Center:* Oscillating WKB density. Sub-barrier states have four turning points and four caustic divergences; above-barrier states have two. *Right:* Resolved density $\rho_{\delta q}(q)$ with the per-orbit Airy density overlaid. The per-well Airy construction has known peaks at the inner turning points (Airy lobe maxima at the well's classical boundary) that clip the panel; the resolved density is everywhere finite and resolves the two-well structure smoothly. For the above-barrier state Airy and the resolved density agree.

smaller. When the orbit reads sub-resolution, the cells return what they can resolve — the floor blob, with Δq and Δp scaled uniformly to saturate $A = \hbar/2$ while preserving the orbit's aspect ratio. An optical microscope reports its aperture, not the size of any sub-resolution feature lying under the lens. The cells report theirs.

The same kinematic root rules out pointwise infinity and pointwise zero. A density at the value ∞ requires concentrating finite probability into zero width: $\delta q \rightarrow 0$, hence $\Delta p \rightarrow \infty$, hence unbounded action. A density at the value 0 requires resolving the field to a single point: $\delta q \rightarrow 0$ again, hence the same unbounded action. The exact zeros of $|\psi_n(q)|^2$ at the wavefunction's nodes are the reciprocal of the WKB infinity at the turning points. Neither survives at any finite resolution. At the cell scale $\delta q = \hbar/\Delta p$, the resolved density takes neither value.

Figure 3 shows the same eigenstates in the wave formalism, with the same kernel widths from the same orbit. The cross-section $W_n(q, 0)$ develops $n + 1$ alternating-sign lobes whose envelope tracks the asymmetric orbit. At resolution δq the cross-section is non-negative [11] and the lobe envelope persists. Husimi, with its fixed coherent-state kernel of width $q_0 = 1$, smooths the envelope into a single peak biased toward the soft wall. The

contrast grows with n as $\delta q/q_0$ shrinks.

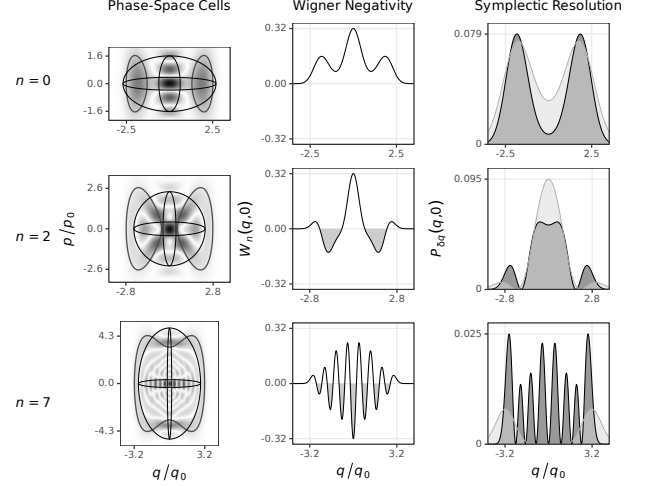


FIG. 5. **Symmetric double well, Wigner.** Same eigenstates as Fig. 4. *Left:* Wigner distribution showing the inter-well interference fringes characteristic of tunneling doublets and the peanut-shaped phase-space orbit of the above-barrier state. *Center:* Cross-section $W_n(q, 0)$. The central peak in the doublet rows is the phase-space signature of coherent superposition between the two wells. *Right:* Cross-section $P_{\delta q}(q, 0)$ with Husimi $Q(q, 0)$ overlaid. The resolved cross-section keeps three peaks — left well, central inter-well coherence, right well; Husimi flattens the central peak.

For the symmetric double well $V(q) = -\mu^2 q^2/2 + \lambda q^4/4$ with $\mu^2 = 4$, $\lambda = 1$, the barrier height is $V_0 = 4$ and the well minima are at $q = \pm 2$. Sub-barrier states form symmetric/antisymmetric tunneling doublets; above-barrier states have a single connected orbit. Figure 4 shows the semiclassical pipeline. The WKB density for sub-barrier states has four caustic divergences, two per well; at resolution δq they become finite peaks. The Airy density, computed per well, places lobe-maxima peaks at every inner turning point — standard behavior of the Airy uniform density at orbit boundaries, applied here to a state whose true wavefunction has its mass at the well minima. The Airy curves clip the panel; the resolved density does not. For the above-barrier state ($n = 7$) the orbit is connected and the two agree.

Figure 5 shows the same eigenstates in the wave formalism. The cross-section $W_n(q, 0)$ for the doublet partners has three peaks: the two well peaks plus a central interference peak that is the phase-space signature of coherent superposition between the wells. At resolution δq the central peak survives. Husimi, with its fixed kernel of width $q_0 = 1$ exceeding the inter-peak separation, flattens the central peak into the wells. The doublet's lowest state is the most stringent test of any phase-space resolution method; the central peak is small in amplitude and narrow in extent, and the cell physics resolves it precisely because $\delta q = \hbar/\Delta p$ is small enough when Δp , set by the

doublet's delocalized covariance, is large.

Tunneling appears in the resolved density. The classical microcanonical density W_{cl} has no support in the double-well barrier, but the kernel's Gaussian tails extend across the barrier from each well. The marginal $\rho_{\delta q}(q)$ is therefore exponentially small but strictly nonzero throughout the barrier interior. The inter-well coupling that conventional treatments add by hand is what the cell physics, asked the right question, reports.

The construction has no free parameters. Δq and Δp come from the orbit; δq and δp from the reciprocal scales; the kernel from the Heisenberg limit. Wigner negativity, semiclassical caustics, and Schrödinger nodes are not three pathologies of three formalisms. They are three artifacts of one unphysical assumption, and they go away when the assumption goes.

All numerical data, figure-generating code, and reproducibility scaffolding for this paper are available at <https://doi.org/10.17605/OSF.IO/EV8H6>.

* bmulloy@umich.edu

- [1] M. V. Berry, Semi-classical mechanics in phase space: A study of Wigner's approach, *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences* **287**, 237 (1977).

- [2] R. E. Langer, On the connection formulas and the solutions of the wave equation, *Physical Review* **51**, 669 (1937).
- [3] E. Wigner, On the quantum correction for thermodynamic equilibrium, *Physical Review* **40**, 749 (1932).
- [4] K. Husimi, Some formal properties of the density matrix, *Proceedings of the Physico-Mathematical Society of Japan. 3rd Series* **22**, 264 (1940).
- [5] H. P. Robertson, The uncertainty principle, *Physical Review* **34**, 163 (1929).
- [6] E. Schrödinger, Zum Heisenbergschen Unschärfeprinzip, *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse* **14**, 296 (1930).
- [9] W. H. Zurek, Sub-Planck structure in phase space and its relevance for quantum decoherence, *Nature* **412**, 712 (2001).
- [10] M. Gromov, Pseudo holomorphic curves in symplectic manifolds, *Inventiones mathematicae* **82**, 307 (1985).
- [11] R. L. Hudson, When is the Wigner quasi-probability density non-negative?, *Reports on Mathematical Physics* **6**, 249 (1974).
- [8] M. A. de Gosson, A geometric picture of the wave function: Fermi's trick (2012), arXiv:1208.0908 [quant-ph].
- [7] M. A. de Gosson, Phase space quantization and the uncertainty principle, *Physics Letters A* **373**, 3896 (2009).
- [] M. de Gosson and C. de Gosson, Symplectic polar duality, quantum blobs, and generalized Gaussians, arXiv preprint (2022), arXiv:2206.06334 [quant-ph].
- [12] D. Brizuela, Classical and quantum behavior of the harmonic and the quartic oscillators, *Physical Review D* **90**, 125018 (2014).